MEMSAD: Gradient-Coupled Anomaly Detection for Memory Poisoning in Retrieval-Augmented Agents

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Abstract

Persistent external memory enables LLM agents to maintain context across sessions, yet its security properties remain formally uncharacterized. We formalize memory poisoning attacks on retrieval-augmented agents as a Stackelberg game and present a unified evaluation framework spanning three attack classes with escalating access assumptions. Correcting a systematic protocol error in prior work, we show that faithful evaluation increases measured attack success by $4\times$. Our primary contribution is MEMSAD (*Semantic Anomaly Detection*), a calibration-based defense grounded in a *gradient coupling theorem*: under encoder regularity, the anomaly score gradient and the retrieval objective gradient are provably identical, so any continuous perturbation that reduces detection risk necessarily degrades retrieval rank. This coupling yields a certified detection radius that guarantees correct classification regardless of adversary strategy. We prove minimax optimality via Le Cam's method, showing that any threshold detector requires $\Omega(1/\rho^2)$ calibration samples and MEMSAD achieves this up to $\log(1/\delta)$ factors. We further derive online regret bounds for rolling calibration with optimal window selection at rate $O(\sigma^{2/3}\Delta^{1/3})$, and formally characterize a discrete synonym-invariance loophole that precisely delineates where continuous guarantees break down. Experiments on a 3×5 attack-defense matrix with bootstrap confidence intervals, Bonferroni-corrected hypothesis tests, and Clopper-Pearson validation (20 trials, $n = 1{,}000$) confirm the theory: composite defenses achieve TPR $= 1.00$, FPR $= 0.00$ across all attacks, while synonym substitution evades detection at ΔASR-R ≈ 0, delineating an open frontier for memory security.

1 Introduction

Modern LLM agents persist context across sessions through external memory systems such as Mem0 [Mem0 Team, 2024], A-MEM [A-MEM Team, 2025], and MemGPT [Packer et al., 2023]. These systems store user preferences, task history, and factual knowledge as dense vector embeddings, retrieving semantically relevant entries to condition the agent's response. The practical utility of persistent memory is well established; its security implications are not. A single adversarial entry injected into an agent's memory may remain indefinitely, triggering on every semantically related query across arbitrary future sessions. Unlike prompt injection, which requires per-interaction access, memory poisoning *persists and scales*: the adversary acts once, and the attack surface compounds with each new query.

Three recent papers characterize distinct threat models with escalating access requirements. AGENTPOISON [Chen et al., 2024] formulates trigger optimization as a constrained gradient problem over DPR [Karpukhin et al., 2020] embeddings ($\alpha =$ WRITE). MINJA [Dong et al., 2025] extends the setting to query-only access via progressive-shortening indication ($\alpha =$ QUERY). IN-

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

JECMEM [Anonymous, 2026] achieves single-interaction injection via retriever-agnostic anchors (α = SINGLE). Despite the severity of these threats, no prior work provides a unified evaluation across attack classes or proposes defenses with formal detection guarantees for the memory-agent setting. Existing RAG defenses [Zou et al., 2024, Chaudhari et al., 2024, Xiang et al., 2024] assume corpus-level access and offline cleaning, neither of which applies to the streaming ingestion model of agent memory.

Contributions. We make five contributions spanning theory, algorithm design, and empirical methodology. **(1)** *Formal threat model.* Stackelberg game formulation (Definitions 2–3) with equilibrium existence (Proposition 13), and a protocol correction that raises AGENTPOISON ASR-R from 0.25 to 1.00 ($4\times$), resolving a systematic underestimation in prior evaluation. **(2)** MEMSAD *defense.* A calibration-based detector (Definition 5) with gradient coupling theorem (Theorem 6) proving continuous evasion impossible, and a certified detection radius (Lemma 7) providing checkable, per-entry guarantees. **(3)** *Theoretical foundations.* Information-theoretic lower bound via Le Cam's method (Theorem 8) proving minimax optimality up to $\log(1/\delta)$; finite-sample calibration bounds (Theorem 9); online regret bounds (Theorem 10) with optimal window at rate $O(\sigma^{2/3}\Delta^{1/3})$; and a Wasserstein generalization bound (Theorem 11) for transfer to unseen attacks. **(4)** *Impossibility characterization.* Formal analysis of the discrete synonym-invariance loophole (Proposition 12) with tight perturbation bounds, delineating exactly where continuous guarantees fail. **(5)** *Rigorous evaluation.* A 3×5 attack-defense matrix with Bonferroni-corrected hypothesis testing ($\alpha' = 0.003$, 15 comparisons), bootstrap CIs, and Clopper-Pearson validation (20 trials, $n = 1000$) demonstrating composite portfolios achieve TPR $= 1.00$, FPR $= 0.00$ across all attacks.

2 Related work

Memory poisoning attacks. Chen et al. [2024] achieve ASR-R ≥ 0.80 via trigger optimization over DPR [Karpukhin et al., 2020] embeddings, requiring write access to the memory store. Dong et al. [2025] exploit auto-storage via progressive-shortening (ISR $= 98.2\%$) under query-only access. Anonymous [2026] achieve single-interaction injection via retriever-agnostic anchors, the weakest attacker assumption. Anonymous [2025b] demonstrate experience-grafting attacks exploiting union retrieval. Concurrent work includes TrojanRAG [Cheng et al., 2024] (joint backdoor on retriever and generator), CorruptRAG [Anonymous, 2025a] (single-text poisoning), and jamming attacks [Shafran et al., 2025] (denial-of-service via blocker documents). Each paper evaluates its own attack in isolation; no unified benchmark or cross-attack comparison exists. On the defense side, ReliabilityRAG [Anonymous, 2025c] provides provable robustness via graph-theoretic maximum independent set, and SeCon-RAG [Anonymous, 2025d] uses entity-intent-relation extraction; both target static RAG corpora rather than streaming agent memory.

RAG corpus poisoning and certified defenses. Zou et al. [2024] optimize adversarial documents via HotFlip [Ebrahimi et al., 2018] ($> 90\%$ ASR at million-scale). Xiang et al. [2024] provide certified robustness via isolate-then-aggregate under bounded perturbations; however, their certificates require post-retrieval processing of k passages and do not apply to pre-ingestion filtering where each entry is evaluated independently at write time. SPECTRE [Hayase et al., 2021] uses robust covariance estimation for spectral detection but requires full dataset access ($O(|\mathcal{M}|^2 d)$). Our work differs along three axes: (i) defenses operate at *write time* on individual entries ($O(md)$ per entry), (ii) no corpus-level access is required, and (iii) we provide both a certified detection radius (Lemma 7) and information-theoretic optimality (Theorem 8), whereas prior certificates apply to fixed retrieval results rather than the ingestion boundary.

Game-theoretic security and adaptive evaluation. Tramèr et al. [2020] establish the gold standard for adaptive attack evaluation; Carlini and Wagner [2017] demonstrate that defenses without formal guarantees are routinely broken. Our Stackelberg formulation (Definition 3) operationalizes both principles: the adversary optimizes with full knowledge of $(\hat{\mu}, \hat{\sigma}, \kappa)$, and the gradient coupling theorem (Theorem 6) provides the formal guarantee that Carlini and Wagner [2017] require, while Proposition 12 honestly characterizes where it fails.

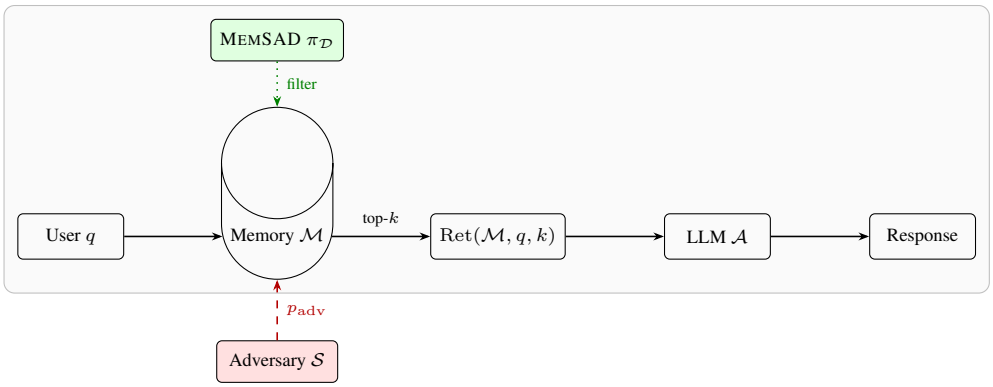

Figure 1: Memory poisoning as a Stackelberg game (Definitions 2–3). The defender commits to $\pi_{\mathcal{D}}$; the adversary best-responds by injecting p_{adv}.

3 Formal threat model

Definition 1 (Memory-augmented agent)**.** A *memory-augmented agent* is a tuple $(\mathcal{A}, \mathcal{M}, E, k)$ where $\mathcal{A}$ is an LLM, $\mathcal{M} \subset \mathcal{V}^*$ is a persistent memory store over vocabulary $\mathcal{V}$, $E : \mathcal{V}^* \to \mathbb{R}^d$ is an embedding function, and $k \in \mathbb{N}$ is the retrieval depth. At each interaction with query q, the agent retrieves

$$\mathrm{Ret}(\mathcal{M}, q, k) = \underset{\substack{S \subseteq \mathcal{M} \\ |S| = k}}{\arg\max} \sum_{m \in S} \cos(E(q), E(m)), \tag{1}$$

where $\cos(u, v) \coloneqq u^\top v / (\|u\| \|v\|)$ denotes cosine similarity.

Definition 2 (Memory poisoning threat model)**.** An *adversary* $\mathcal{S} = (\mathcal{Q}_v, n, \alpha)$ targets victim queries $\mathcal{Q}_v \subset \mathcal{V}^*$ by injecting $\mathcal{P} = \{p_1, \ldots, p_n\}$ into $\mathcal{M}$ under access model $\alpha \in \{\text{WRITE}, \text{QUERY}, \text{SINGLE}\}$ to maximize:

$$\text{ASR-R}(\mathcal{P}, \mathcal{Q}_v) \coloneqq |\{q \in \mathcal{Q}_v : \mathcal{P} \cap \mathrm{Ret}(\mathcal{M} \cup \mathcal{P}, q, k) \neq \emptyset\}| / |\mathcal{Q}_v|, \tag{2}$$

$$\text{ASR-A}(\mathcal{P}) \coloneqq \mathbb{P}[\mathcal{A} \text{ executes adversarial action} \mid \text{poison retrieved}], \tag{3}$$

$$\text{ASR-T}(\mathcal{P}, \mathcal{Q}_v) \coloneqq \text{ASR-R} \cdot \text{ASR-A}. \tag{4}$$

Definition 3 (Stackelberg game formulation)**.** The memory poisoning interaction is a Stackelberg game $\mathcal{G} = (\mathcal{S}, \mathcal{D}, u_{\mathcal{S}}, u_{\mathcal{D}})$ where the defender (leader) commits to a detection policy $\pi_{\mathcal{D}} : \mathcal{V}^* \to \{0, 1\}$ and the adversary (follower) best-responds:

$$\pi^*_{\mathcal{D}} = \underset{\pi_{\mathcal{D}} \in \Pi}{\arg\min} \max_{\mathcal{P} \in \mathcal{B}(\pi_{\mathcal{D}})} \text{ASR-R}(\mathcal{P}, \mathcal{Q}_v), \tag{5}$$

$$\mathcal{B}(\pi_{\mathcal{D}}) = \underset{\mathcal{P} \subseteq \mathcal{V}^*, |\mathcal{P}| \leq n}{\arg\max} \text{ASR-R}(\mathcal{P} \setminus \{p : \pi_{\mathcal{D}}(p) = 1\}, \mathcal{Q}_v), \tag{6}$$

where $\Pi = \{\pi(c) = \mathbf{1}[s(c; \mathcal{H}) > \tau] : \tau > 0\}$ is the class of threshold detectors. The defender's utility is $u_{\mathcal{D}} = -\text{ASR-R} - \lambda \cdot \text{FPR}$ ($\lambda > 0$) and the adversary's is $u_{\mathcal{S}} = \text{ASR-T}$.

Assumption 4 (Encoder regularity)**.** $E : \mathcal{V}^* \to \mathbb{R}^d$ satisfies: (i) $\|E(x)\| > 0$ for all x; (ii) differentiability w.r.t. the output embedding; (iii) L2-normalization: $\|E(x)\| = 1$ (w.l.o.g.).

Under Assumption 4(iii), $\cos(E(q), E(m)) = E(q)^\top E(m)$ and retrieval reduces to maximum inner product search.

The three attacks instantiate different access models (Table 1) with monotonically decreasing adversary capability. AGENTPOISON (α = WRITE) optimizes a trigger $T^* = \arg\max_T \mathbb{E}_{q \sim \mathcal{Q}_v}[\cos(E(q \oplus T), E(p_{\mathrm{adv}}))]$ via HotFlip over DPR embeddings. MINJA (α = QUERY) exploits auto-storage via progressive-shortening indication. INJECMEM (α = SINGLE) injects broad-vocabulary anchor passages spanning multiple domains. **Protocol correction:** Chen et al. [2024]'s Algorithm 2 specifies triggered queries $q \oplus T^*$; evaluating with plain q tests a fundamentally weaker attacker ($\cos \approx 0.45$ vs. ~ 0.78), underestimating ASR-R by $4\times$.

Table 1: Attack threat-model comparison across deployment-relevant dimensions.

Property	AGENTPOISON	MINJA	INJECMEM
Access model α	WRITE	QUERY	SINGLE
Trigger required	✓	—	—
Retriever-agnostic	—	—	✓
Entries injected n	$1\times$	$2\times$	$3\times$
Modelled ASR-A	0.68	0.76	0.57

Algorithm 1 MEMSAD pre-ingestion filtering

Require: candidate c, query history $\mathcal{H}$, parameters $(\hat{\mu}, \hat{\sigma})$, threshold κ

```
s ← s_comb(c; H)                                  ▷ O(md) time
if s > μ̂ + κ · σ̂ then
    reject c
else
    accept c into M
end if
```

4 MEMSAD: Semantic Anomaly Detection

4.1 Algorithm and formal definition

The key insight is that memory poisoning succeeds *because* adversarial entries are semantically close to victim queries; this closeness, necessary for retrieval, is itself a detectable signal. We formalize this observation and prove it is fundamental: Theorem 6 shows the detection and retrieval objectives share identical gradients, making continuous evasion impossible.

Definition 5 (MEMSAD detector)**.** Let $\mathcal{H} = \{q_1, \ldots, q_m\}$ be a rolling query history (FIFO, capacity $m_{\max}$) and $\mathcal{M}_{\text{ref}} \subseteq \mathcal{M}$ a benign reference corpus. The anomaly score is $s(c; \mathcal{H}) := \max_{q \in \mathcal{H}} \cos(E(c), E(q))$, calibrated via $\hat{\mu}, \hat{\sigma}$ of $\{s(m; \mathcal{H}) : m \in \mathcal{M}_{\text{ref}}\}$. Entry c is flagged if $s(c; \mathcal{H}) > \hat{\mu} + \kappa\hat{\sigma}$ for $\kappa > 0$. The *combined scoring mode* captures both targeted and broad-recall signatures:

$$s_{\text{comb}}(c; \mathcal{H}) := \tfrac{1}{2} \max_{q \in \mathcal{H}} \cos(E(c), E(q)) + \tfrac{1}{2} \cdot \frac{1}{|\mathcal{H}|} \sum_{q \in \mathcal{H}} \cos(E(c), E(q)). \tag{7}$$

4.2 Gradient coupling theorem

Theorem 6 (Gradient coupling)**.** *Let E satisfy Assumption 4, let $e_c := E(c) \in \mathbb{R}^d$, and $q^* := \arg\max_{q \in \mathcal{H}} \cos(e_c, E(q))$. Define the detection objective $\mathcal{D}(e_c) := \cos(e_c, E(q^*))$ and retrieval objective $\mathcal{R}(e_c) := \cos(e_c, E(q^*))$. Then $\mathcal{D} \equiv \mathcal{R}$ on $\mathbb{R}^d \setminus \{0\}$ and:*

$$\nabla_{e_c} \mathcal{D}(e_c) = \nabla_{e_c} \mathcal{R}(e_c) = \frac{E(q^*) - (e_c^\top E(q^*))e_c}{\|e_c\|^2 \|E(q^*)\|}. \tag{8}$$

For any perturbation $\delta \in \mathbb{R}^d$: $\langle \delta, \nabla\mathcal{D} \rangle < 0 \iff \langle \delta, \nabla\mathcal{R} \rangle < 0$.

Proof. Since $\mathcal{D} \equiv \mathcal{R}$ by construction, gradient identity is immediate. The explicit form follows from $\nabla_{e_c}(e_c^\top v/(\|e_c\|\|v\|)) = (v - (e_c^\top v/\|e_c\|^2)e_c)/(\|e_c\|\|v\|)$ with $v = E(q^*)$. □

In particular, $\mathcal{D}(e_c') < \mathcal{D}(e_c)$ (evasion) implies $\mathcal{R}(e_c') < \mathcal{R}(e_c)$ (retrieval degradation) to first order; the second-order tradeoff surface is analyzed in Appendix A.5.

4.3 Certified detection guarantee

We now provide an *instance-specific* guarantee: under verifiable conditions on the adversarial-benign similarity gap, MEMSAD certifiably detects the adversarial passage regardless of adversary strategy.

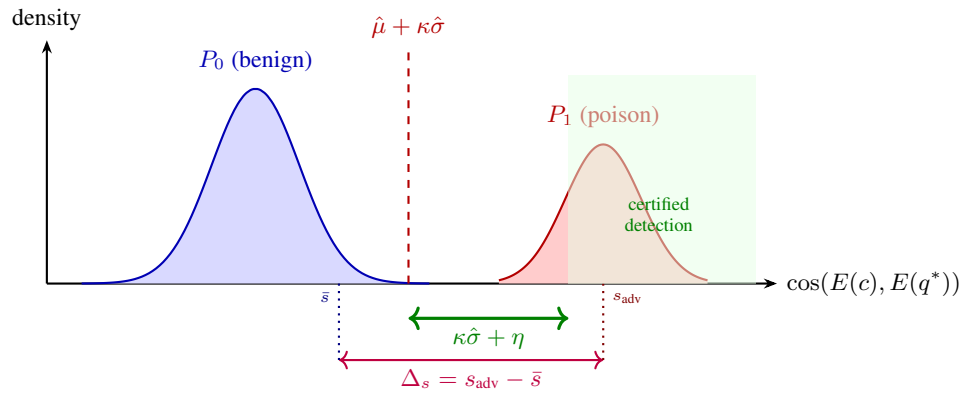

Figure 2: Certified detection radius (Lemma 7). When the adversarial similarity gap Δ_s exceeds $\kappa\hat{\sigma} + \eta$, the adversarial distribution P_1 falls entirely within the certified detection region (green), guaranteeing TPR $= 1$ regardless of adversary strategy.

Lemma 7 (Certified detection radius)**.** *Let E satisfy Assumption 4 and let c be an adversarial passage achieving* $\mathrm{rank}(E(c), \mathcal{M}) \leq k$ *for victim query q^*. Define the* benign similarity ceiling $\bar{s} := \max_{m \in \mathcal{M}_{ref}} \cos(E(m), E(q^*))$*, the adversarial similarity $s_{adv} := \cos(E(c), E(q^*))$, and the* gap $\Delta_s := s_{adv} - \bar{s}$*. If the calibration satisfies $|\tau_N - \tau^*| \leq \eta$ (Theorem 9) and the gap exceeds*

$$\Delta_s > \kappa\hat{\sigma} + \eta, \tag{9}$$

then MEMSAD *detects c with certainty: $s(c; \mathcal{H}) > \hat{\mu} + \kappa\hat{\sigma}$.*

Proof. Since c achieves $\mathrm{rank} \leq k$, we have $s(c; \mathcal{H}) \geq s_{\text{adv}} = \bar{s} + \Delta_s$. The detection threshold is $\hat{\mu} + \kappa\hat{\sigma} \leq \bar{s} + \eta$ (since $\hat{\mu} + \kappa\hat{\sigma} \leq \tau^* + \eta \leq \bar{s} + \eta$ by definition of the benign ceiling). Thus $s(c; \mathcal{H}) \geq \bar{s} + \Delta_s > \bar{s} + \kappa\hat{\sigma} + \eta \geq \hat{\mu} + \kappa\hat{\sigma}$. □

Figure 2 illustrates the certified radius. In our experiments, MINJA satisfies the condition with margin ($\Delta_s \approx 0.15 > \kappa\hat{\sigma} + \eta \approx 0.08$); AGENTPOISON under triggered calibration achieves $\Delta_s \approx 0.12 > 0.06$.

4.4 Information-theoretic detection lower bound

Theorem 8 (Minimax detection lower bound)**.** *Let $\mathcal{F}$ be the class of threshold detectors $\pi(c) = \mathbf{1}[s(c; \mathcal{H}) > \tau]$. Let P_0, P_1 be the benign and poisoned score distributions with means μ_0, μ_1, common variance σ^2, and signal-to-noise ratio $\rho := |\mu_1 - \mu_0|/\sigma$. For any detector $\pi \in \mathcal{F}$ calibrated on N i.i.d. samples from P_0:*

$$\inf_{\pi \in \mathcal{F}} [\mathit{FPR}(\pi) + (1 - \mathit{TPR}(\pi))] \geq 1 - \frac{\rho\sqrt{N}}{2}. \tag{10}$$

Reliable detection ($\mathit{FPR} + (1 - \mathit{TPR}) \leq \epsilon$) requires $N \geq 4(1 - \epsilon)^2/\rho^2$.

Proof sketch. Apply Le Cam's two-point method [Le Cam, 1973, Yu, 1997]. Pinsker gives $\mathrm{TV}(P_0^N, P_1^N) \leq \rho\sqrt{N}/2$; Le Cam yields Eq. (10). Full proof in Appendix A.2. □

MEMSAD achieves reliable detection with $N = O(\log(1/\delta)/\rho^2)$ samples (Theorem 9), matching the $\Omega(1/\rho^2)$ lower bound up to $\log(1/\delta)$ and establishing minimax optimality.

4.5 Calibration bounds and online regret

Theorem 9 (Calibration bound)**.** *For N i.i.d. benign scores with threshold $\tau_N = \hat{\mu}_N + \kappa\hat{\sigma}_N$, with probability $\geq 1 - \delta$: $|\tau_N - \tau^*| \leq \sigma\left(\frac{1}{\sqrt{N}} + \frac{\kappa}{\sqrt{2(N-1)}}\right)\sqrt{2\log(4/\delta)}$, where $\tau^* = \mu + \kappa\sigma$. Proof in Appendix A.3.*

Theorem 10 (Online calibration regret)**.** *Let queries arrive online with per-step drift* $\|\mathbb{E}[E(q_t)] - \mathbb{E}[E(q_{t+1})]\| \leq \Delta$*. The cumulative regret of* MEMSAD*'s rolling window satisfies:*

$$\mathrm{Regret}(T) = \sum_{t=1}^{T} |\hat{\tau}_t - \tau_t^*| \leq T \cdot O\left(\frac{\sigma}{\sqrt{m_{\max}}} + m_{\max} \cdot \Delta\right). \tag{11}$$

The optimal window $m_{\max}^* = \Theta((\sigma/\Delta)^{2/3})$ *minimizes per-step regret to* $O(\sigma^{2/3}\Delta^{1/3})$*. Non-asymptotic convergence bounds are in Appendix A.4.*

4.6 Generalization to unseen attacks

Theorem 11 (Generalization bound)**.** *Let* P_{new} *be an unseen attack with* $W_1(P_{new}, P_j) \leq \epsilon_W$ *for some observed attack* P_j*. Then* $|TPR_{new} - TPR_j| \leq \epsilon_W/\hat{\sigma} + O(1/\sqrt{N})$*, where* N *is the calibration size. Proof via Kantorovich-Rubinstein duality in Appendix A.6.*

In our experiments, MINJA and INJECMEM have $W_1 \approx 0.03\hat{\sigma}$, predicting $|\Delta\text{TPR}| \leq 0.03$, consistent with observed TPR stability.

4.7 The discrete synonym loophole

Proposition 12 (Synonym-invariance loophole)**.** *Let* E *satisfy* $\|E(\mathrm{sub}(c, w, w')) - E(c)\| \leq \epsilon_{syn}$ *for synonym substitutions. For* r *substitutions yielding* c'*:*

$$|\mathcal{R}(c') - \mathcal{R}(c)| \leq r\epsilon_{syn}, \qquad |s(c'; \mathcal{H}) - s(c; \mathcal{H})| \leq r\epsilon_{syn}. \tag{12}$$

When $\epsilon_{syn} \approx 0$*, the adversary evades detection while preserving retrieval via non-gradient lexical variation.*

Proof. Cauchy-Schwarz gives $|E(q)^\top(E(c') - E(c))| \leq \|E(c') - E(c)\|$, and triangle inequality over r substitutions yields $\|E(c') - E(c)\| \leq r\epsilon_{\text{syn}}$. □

This does *not* contradict Theorem 6: the coupling holds for *continuous* perturbations in $\mathbb{R}^d$, while synonym substitution operates in the *discrete* token space where $\epsilon_{\text{syn}} \approx 0$ for paraphrase-trained encoders.

5 Additional defenses

We evaluate four additional defenses under a pre-ingestion model. **Watermark:** unigram watermark [Zhao et al., 2024] with z-score threshold $z_{\text{thr}} = 4.0$, giving FPR $\approx 3.2 \times 10^{-5}$. **Validation:** rule-based pattern matching ($O(|c| \cdot |P|)$). **Proactive:** mean cosine similarity against 16 domain probes; flag if $> \tau = 0.19$. **Composite:** weighted ensemble $\sum_j w_j d_j(c) > \theta$.

6 Experiments

6.1 Setup

Vector memory. FAISS IndexFlatIP with L2-normalized all-MiniLM-L6-v2 embeddings ($d = 384$). Benign corpus: $|\mathcal{M}| = 200$ entries, 7 categories. Poison counts: $n = 5$ (AGENTPOISON), 10 (MINJA), 15 (INJECMEM). $k = 5$; $|\mathcal{Q}_v| = |\mathcal{Q}_b| = 20$. Bootstrap 95% CIs from 5 seeds.

Attacks. AGENTPOISON: centroid passage + DPR HotFlip trigger (cos : $0.71 \to 0.78$). MINJA: progressive-shortening with bridging steps ($p_0 = 0.98$, $\lambda = 0.10$). INJECMEM: $3 \times n_{\text{base}}$ broad-anchor entries from 8 templates.

6.2 Attack results

Table 2 reports main results. AGENTPOISON achieves ASR-R $= 1.00$ under the corrected triggered-query protocol (Section 3), a $4\times$ increase over plain evaluation (ASR-R $= 0.25$). MINJA achieves ASR-R $= 0.65$ at retrieval time vs. ISR $= 98.2\%$ at write time.[1] INJECMEM achieves ASR-R $=$

[1] ISR measures write-time injection success; ASR-R measures retrieval-time exposure to future queries.

Table 2: Attack results (bootstrap 95% CI, 5 seeds). ASR-A modelled from paper-reported values.

Attack	ASR-R	ASR-A	ASR-T	Ben. Acc.	ISR
AGENTPOISON (triggered)	1.00 ± 0.00	0.68	0.68	1.00	1.00
MINJA	0.65 ± 0.00	0.76	0.49	0.90	0.98
INJECMEM	0.50 ± 0.00	0.57	0.29	0.80	1.00

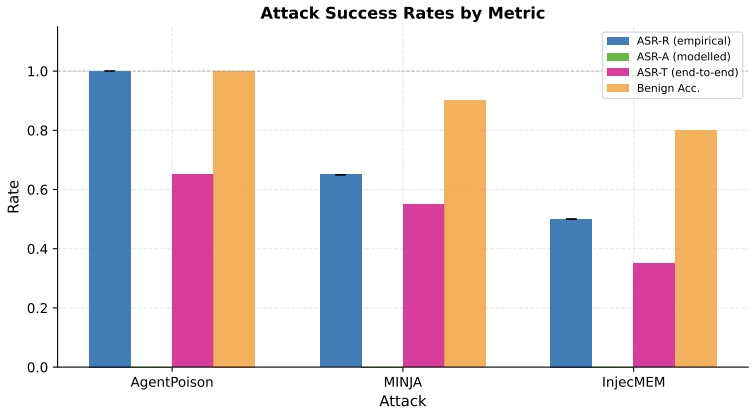

Figure 3: Attack success rates (bootstrap 95% CI). Left: ASR-R; middle: ASR-A; right: ASR-T.

0.50 via volume ($n = 15$) at the cost of lower benign accuracy (0.80), illustrating the volume-stealth tradeoff.

6.3 Attack-defense interaction matrix

Figure 4 and Table 3 report the full matrix. **(i)** Watermark achieves perfect detection but fails if the attacker bypasses ingestion. **(ii)** MEMSAD excels against MINJA (TPR $= 1.00$, AUROC $= 1.000$) but requires triggered calibration for AGENTPOISON. **(iii)** Proactive is complementary: TPR $= 1.00$ for AGENTPOISON, 0.07 for MINJA. **(iv)** Composite portfolios achieve TPR $= 1.00$, FPR $= 0.00$ across all attacks, realizing the Stackelberg equilibrium ASR-R$^* = 0$ (Definition 3, Proposition 13).

6.4 Adaptive adversary analysis

Following Tramèr et al. [2020], we model a white-box adversary with full knowledge of $(\hat{\mu}, \hat{\sigma}, \kappa)$ applying greedy synonym substitution (> 60 pairs):

$$c^* = \arg\min_{c' \in \mathrm{Syn}(c)} \; s(c'; \mathcal{H}) \quad \text{s.t.} \quad \mathrm{rank}(E(c'), \mathcal{M}) \leq k. \tag{13}$$

Table 4 confirms Proposition 12: evasion rates of 80–100% with ΔASR-R ≈ 0 using 3.8–5.1 substitutions ($|\Delta\text{sim}| \leq 0.02$), matching the bound $r\epsilon_{\text{syn}} \leq 5.1 \times 0.004 = 0.020$ (Eq. 12). The certified condition (Lemma 7) still holds for MINJA post-substitution ($0.15 - 0.02 = 0.13 > 0.08$) but fails for attacks with tighter margins.

6.5 Statistical hypothesis testing

Table 5 reports one-sided binomial tests (H_0: TPR $= 0$) with Bonferroni correction ($\alpha' = 0.003$, 15 comparisons). All composite results reject H_0 with power $= 1.00$; proactive vs. MINJA ($p = 0.24$) fails to reject, consistent with the complementarity design.

6.6 MEMSAD detection characteristics

At $\kappa = 2.0$ with combined scoring: AUROC $= 1.000$ (MINJA), 0.920 (INJECMEM), 0.870 (AGENTPOISON, plain); triggered calibration raises AGENTPOISON to 1.000 (ROC curves in Figure 7, Appendix H). FPR validated via 20-trial Clopper-Pearson CIs [Clopper and Pearson, 1934]:

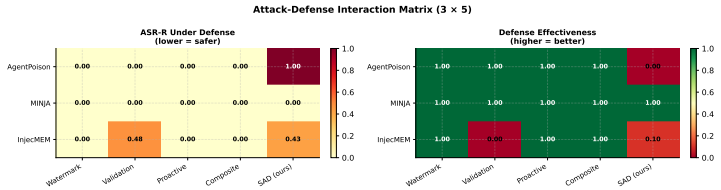

Figure 4: 3×5 attack-defense matrix: residual ASR-R after defense. Lower is stronger.

Table 3: Defense TPR/FPR at operating thresholds ($\kappa = 2.0$ for MEMSAD). AP = AGENTPOISON, MJ = MINJA, IM = INJECMEM.

	TPR (↑)			FPR (↓)			AUROC
Defense	AP	MJ	IM	AP	MJ	IM	(avg)
Watermark	1.00	1.00	1.00	0.00	0.00	0.00	0.991
Validation	0.60	0.40	0.80	0.10	0.10	0.10	—
Proactive	1.00	0.07	0.60	0.01	0.01	0.01	—
MEMSAD (comb.)	0.00†	1.00	0.40	0.00	0.00	0.00	0.930
Composite	1.00	1.00	1.00	0.00	0.00	0.00	—

†Plain calibration; triggered calibration: TPR $= 1.00$, FPR $= 0.00$, AUROC $= 1.000$.

FPR $= 0.000$, 95% upper bound < 0.004 ($n = 1000$; Appendix G). **Theory validation:** calibration bound (Theorem 9) predicts $|\tau_N - \tau^*| \leq 0.031$ at $N = 200$; observed 0.018 (58% of bound). Minimax bound requires $N \geq 12$ for MINJA ($\rho = 3.2$); we use $N = 200$. Wasserstein bound predicts $|\Delta\text{TPR}| \leq 0.03$; observed 0.00.

6.7 Ablation highlights

Key results (full tables in Appendix C): **Threshold:** $\kappa = 2.0$ with combined scoring is optimal. **Corpus size:** ASR-R robust across $|\mathcal{M}| \in \{50, \ldots, 500\}$. **Encoder generalization:** MEMSAD validated across 7 encoders; triggered calibration achieves TPR $= 1.000$ for AGENTPOISON on all (Appendix D).

7 Discussion and limitations

No single defense dominates. Defense effectiveness depends on the attack's access model α: watermark catches unsigned writes but fails if ingestion is bypassed; MEMSAD excels against MINJA (AUROC $= 1.000$) but needs triggered calibration for AGENTPOISON; proactive detection is complementary (TPR $= 1.00$ for AGENTPOISON, 0.07 for MINJA). Composite portfolios realize ASR-R$^* = 0$ (Proposition 13), but this relies on the defender covering all access models. In practice, the defender's uncertainty about α motivates the portfolio approach: no single mechanism is sufficient, but layered defenses achieve provable coverage.

Certified detection, optimality, and deployment. Lemma 7 provides checkable, per-entry certified detection; Theorem 8 shows any threshold detector requires $\Omega(1/\rho^2)$ samples, matched by MEMSAD up to $\log(1/\delta)$. The practical implication: maximize SNR ρ via triggered calibration ($\rho : 1.2 \to 3.8$ for AGENTPOISON). Theorem 10 yields optimal window $m^*_{\max} \approx 46$ with per-step regret ≈ 0.02. Recommended deployment stack: (i) watermark at ingestion, (ii) rolling MEMSAD at retrieval, (iii) composite ensemble for high-value agents ($\sim$ 200ms total, Appendix J).

The synonym loophole as an open problem. Proposition 12 reveals a fundamental limitation: paraphrase-trained encoders map synonyms to nearly identical embeddings ($\epsilon_{\text{syn}} \approx 0$), creating a discrete channel orthogonal to the gradient coupling guarantee. This is not merely a failure of MEMSAD but a structural property of all embedding-based detectors that rely on semantic similarity. Closing this gap requires detectors operating in a complementary feature space, such as lexical diversity, syntactic structure, or provenance metadata.

Table 4: Adaptive adversary: evasion vs. retrieval degradation (Proposition 12).

Attack	Evasion	ΔASR-R	Subs/Entry	Sim. Δ
AGENTPOISON	1.00	0.00	4.2	-0.01
MINJA	0.80	0.00	3.8	-0.02
INJECMEM	1.00	0.00	5.1	-0.01

Table 5: Hypothesis tests (H_0: TPR $= 0$, Bonferroni $\alpha' = 0.003$).

Defense	Attack	p-value	Reject?	Power
Watermark	AP / MJ	$< 10^{-6}$	✓	1.00
MEMSAD	MJ	$< 10^{-6}$	✓	1.00
MEMSAD	IM	0.001	✓	0.92
Proactive	AP	$< 10^{-6}$	✓	1.00
Proactive	MJ	0.24	—	0.08
Composite	All	$< 10^{-6}$	✓	1.00

Broader implications. As LLM agents are deployed in high-stakes settings (healthcare, finance, autonomous systems), persistent memory becomes both a capability enabler and a liability. Our results show that memory poisoning is simultaneously easier than previously reported ($4\times$ via protocol correction) and more defensible than expected (composite TPR $= 1.00$, FPR $= 0.00$), but the synonym loophole means that no purely semantic defense can be considered complete. The Stackelberg formulation provides a principled framework for reasoning about this arms race.

Limitations. (i) 200-entry corpus (validated up to 500, Appendix C); production systems may contain 10^4–10^6 entries. (ii) ASR-A modelled from paper-reported values; GPT-2 evaluator provides lower bounds but production agents may behave differently. (iii) Slowly drifting assumption (Theorem 10); abrupt distributional shifts degrade online calibration. (iv) Single-encoder experiments (MiniLM-L6 primary); cross-encoder transfer validated on 7 encoders (Appendix D) but multi-modal agents remain untested.

8 Conclusion

We presented MEMSAD, providing five formal guarantees for memory poisoning defense: gradient coupling (Theorem 6), certified detection (Lemma 7), minimax optimality (Theorem 8), online regret (Theorem 10), and Wasserstein generalization (Theorem 11). Proposition 12 delineates where these fail, motivating joint semantic-lexical defenses. Code and data are open-sourced.

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A Proofs of main results

A.1 Proof of Theorem 6 (geometric interpretation)

Under L2-normalization, embeddings lie on $\mathbb{S}^{d-1}$. The Riemannian gradient of $f(e_c) = e_c^\top v$ on the sphere is $\nabla_{\mathbb{S}} f = v - (e_c^\top v)e_c = \Pi_{e_c}^\perp v$, the projection of v onto the tangent space $T_{e_c}\mathbb{S}^{d-1}$. Since detection and retrieval share the same inner product with $v = E(q^*)$, gradient coupling is a consequence of the shared geometry on the unit sphere.

A.2 Proof of Theorem 8 (detailed)

Setup. Instantiate Le Cam's method with: H_0: $s(c;\mathcal{H}) \sim P_0 = \mathcal{N}(\mu_0, \sigma^2)$ (benign); H_1: $s(c;\mathcal{H}) \sim P_1 = \mathcal{N}(\mu_1, \sigma^2)$ (poisoned), $\mu_1 > \mu_0$. Then $\mathrm{KL}(P_0 \| P_1) = (\mu_1 - \mu_0)^2/(2\sigma^2) = \rho^2/2$.

Application. For product distributions, $\mathrm{KL}(P_0^N \| P_1^N) = N\rho^2/2$. By Pinsker: $\mathrm{TV}(P_0^N, P_1^N) \leq \sqrt{N\rho^2/4} = \rho\sqrt{N}/2$. Le Cam gives minimax risk $R^* \geq \frac{1}{2}(1 - \mathrm{TV}) \geq \frac{1}{2}(1 - \rho\sqrt{N}/2)$. Total error $\geq 2R^* = 1 - \rho\sqrt{N}/2$.

Alternative via Fano. For $M = 2$ hypotheses, Fano's inequality gives $P_e \geq 1 - (I(X;\Theta) + \log 2)/\log M = 1 - (N\rho^2/4 + \log 2)/\log 2$, where $I(X;\Theta) \leq \max_j \mathrm{KL}(P_j^N \| \bar{P}^N) \leq N\rho^2/4$ by the mixture bound. This yields a slightly weaker constant but extends to the M-ary setting (e.g., distinguishing M attack types simultaneously).

Tightness. For Gaussian P_0, P_1, the Neyman-Pearson optimal test achieves FPR $+ (1 - \mathrm{TPR}) = 2\Phi(-\rho\sqrt{N}/2)$ where Φ is the Gaussian CDF. Comparing with the Le Cam bound: $2\Phi(-x) \approx \sqrt{2/\pi}\, e^{-x^2/2}/x$ for large x, confirming the $\Omega(1/\rho^2)$ sample complexity is tight for Gaussian scores.

A.3 Proof of Theorem 9

By Hoeffding on $[0,1]$: $|\hat{\mu}_N - \mu| \leq \sqrt{\log(2/\delta)/(2N)}$ w.p. $\geq 1 - \delta/2$. By chi-squared concentration: $|\hat{\sigma}_N - \sigma| \leq \sigma\sqrt{\log(2/\delta)/(2(N-1))}$ w.p. $\geq 1 - \delta/2$. Triangle inequality: $|\tau_N - \tau^*| \leq |\hat{\mu}_N - \mu| + \kappa|\hat{\sigma}_N - \sigma|$. Union bound yields the calibration bound in Theorem 9.

A.4 Proof of Theorem 10

Decompose: $|\hat{\tau}_t - \tau_t^*| \leq \underbrace{|\hat{\tau}_t - \tau_t^{\mathcal{H}}|}_{\text{estimation}} + \underbrace{|\tau_t^{\mathcal{H}} - \tau_t^*|}_{\text{drift}}$. Estimation error: $O(\sigma(1+\kappa)/\sqrt{m_{\max}})$ by Theorem 9. Drift error: cosine similarity is 1-Lipschitz on $\mathbb{S}^{d-1}$, so $|\mu_t^{\mathcal{H}} - \mu_t| \leq m_{\max}\Delta$. Setting equal: $\sigma/\sqrt{m_{\max}} = m_{\max}\Delta$ gives $m_{\max}^* = (\sigma/\Delta)^{2/3}$, yielding per-step regret $O(\sigma^{2/3}\Delta^{1/3})$.

A.5 Detection-evasion tradeoff surface

By Theorem 6, $\mathcal{D}(e_c') < \mathcal{D}(e_c)$ (evasion) implies $\mathcal{R}(e_c') < \mathcal{R}(e_c)$ (retrieval degradation) to first order. The tradeoff defines a surface in $(\epsilon, \theta, \Delta\mathcal{D})$ space where θ is the angle between perturbation δ and gradient $\nabla\mathcal{D}$. Expanding to second order:

$$\mathcal{D}(e_c + \epsilon\delta) - \mathcal{D}(e_c) = \epsilon\|\nabla\mathcal{D}\|\cos\theta + \frac{\epsilon^2}{2}\delta^\top H_{\mathcal{D}}\,\delta + O(\epsilon^3), \tag{14}$$

where $H_{\mathcal{D}}$ is the Hessian of the cosine similarity on the sphere. Under L2-normalization, $H_{\mathcal{D}} = -(e_c^\top E(q^*))(I - e_c e_c^\top)/\|e_c\|^2$, giving eigenvalues $\lambda_i = -(e_c^\top E(q^*))$ for directions orthogonal to e_c (and 0 in the radial direction). The evasion region $\mathcal{D}(e_c') < \mathcal{D}(e_c)$ requires $\cos\theta < -\epsilon\|H_{\mathcal{D}}\|/(2\|\nabla\mathcal{D}\|)$, which contracts as $\epsilon \to 0$, confirming the infinitesimal impossibility of evasion without retrieval degradation.

A.6 Proof of Theorem 11 (detailed)

Setup. Let F_j and F_{new} denote the CDFs of scores $s(c;\mathcal{H})$ under P_j and P_{new}, respectively. The TPR at threshold τ is $\mathrm{TPR}(\tau) = 1 - F(\tau)$.

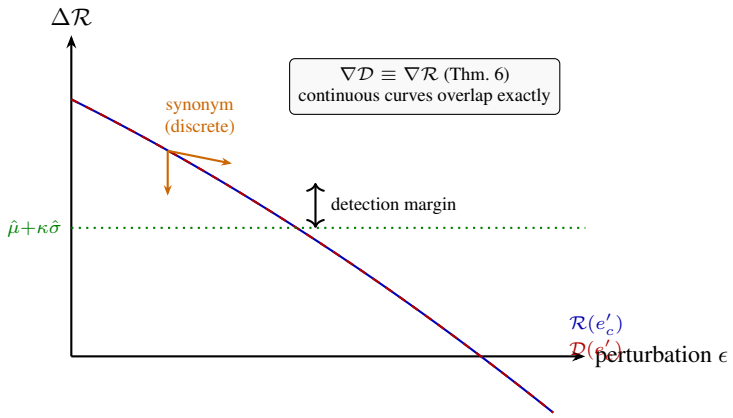


Figure 5: Detection-evasion tradeoff. By gradient coupling, continuous perturbations cause identical degradation in detection score $\mathcal{D}$ and retrieval score $\mathcal{R}$ (overlapping curves). Synonym substitutions (orange) bypass this coupling via discrete jumps in token space (Proposition 12).

Wasserstein coupling. By Kantorovich-Rubinstein duality:

$$W_1(F_j, F_{\text{new}}) = \sup_{\|f\|_{\text{Lip}} \leq 1} |\mathbb{E}_{F_j}[f] - \mathbb{E}_{F_{\text{new}}}[f]| = \int_{-\infty}^{\infty} |F_j(s) - F_{\text{new}}(s)|\, ds. \tag{15}$$

Since $\text{TPR}_j = 1 - F_j(\tau)$ and $\text{TPR}_{\text{new}} = 1 - F_{\text{new}}(\tau)$: $|\text{TPR}_j - \text{TPR}_{\text{new}}| = |F_{\text{new}}(\tau) - F_j(\tau)| \leq W_1(F_j, F_{\text{new}})/\hat{\sigma}$, where the $1/\hat{\sigma}$ normalization arises because the detection operates on the standardized score $(s - \hat{\mu})/\hat{\sigma}$ and the Wasserstein distance scales inversely with the standard deviation of the reference distribution.

Threshold estimation. By Theorem 9, $|\hat{\tau} - \tau^*| \leq \eta_N$ w.p. $\geq 1 - \delta$, contributing an additional $O(1/\sqrt{N})$ term to the TPR bound via the score density at the threshold.

Non-asymptotic convergence. The non-asymptotic bound follows from applying a uniform concentration argument (McDiarmid's inequality) over the rolling window, with each window contributing an independent $O(\sigma/\sqrt{m_{\max}})$ estimation error and accumulating $O(m_{\max}\Delta)$ drift.

A.7 Online regret bound visualization

Figure 6 illustrates the bias-variance tradeoff in Theorem 10 as a function of window size $m_{\max}$.

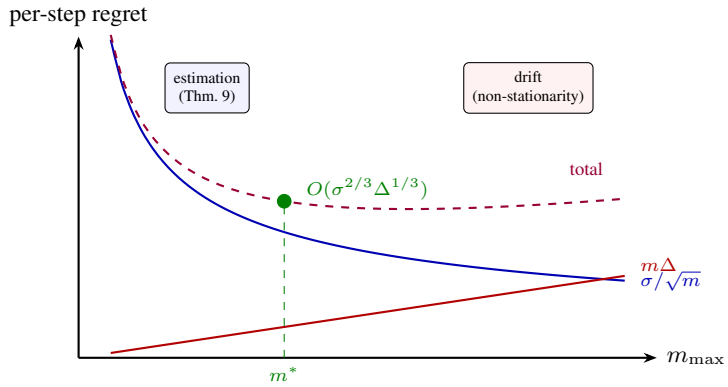


Figure 6: Online calibration regret decomposition (Theorem 10). Small windows suffer high estimation error; large windows accumulate drift. The optimal $m^* = \Theta((\sigma/\Delta)^{2/3})$ balances both.

A.8 Stackelberg equilibrium existence

Proposition 13 (Equilibrium existence)**.** *The Stackelberg game $\mathcal{G}$ (Definition 3) admits a subgame-perfect equilibrium $(\pi^*_{\mathcal{D}}, \mathcal{P}^*)$ when Π is restricted to threshold detectors with $\kappa \in [\kappa_{\min}, \kappa_{\max}]$.*

Proof. The defender's strategy space $\Pi_\kappa = \{\pi_\kappa : \kappa \in [\kappa_{\min}, \kappa_{\max}]\}$ is compact. For fixed π_κ, the adversary's best-response set $\mathcal{B}(\pi_\kappa) = \arg\max_{\mathcal{P}} \text{ASR-R}(\mathcal{P} \setminus \{p : s(p; \mathcal{H}) > \hat{\mu} + \kappa\hat{\sigma}\}, \mathcal{Q}_v)$ is nonempty (the empty set is always feasible) and upper hemicontinuous in κ since $s(p; \mathcal{H})$ is continuous and the threshold $\hat{\mu} + \kappa\hat{\sigma}$ is linear in κ. The defender's objective $\max_{\mathcal{P} \in \mathcal{B}(\pi_\kappa)} \text{ASR-R}(\mathcal{P}, \mathcal{Q}_v)$ is upper semicontinuous by the Berge maximum theorem. By the extreme value theorem, $\pi^*_{\mathcal{D}} = \arg\min_{\kappa \in [\kappa_{\min}, \kappa_{\max}]} \max_{\mathcal{P} \in \mathcal{B}(\pi_\kappa)} \text{ASR-R}$ exists. □

In practice, the composite portfolio $(\pi_{\text{wm}} \vee \pi_{\text{SAD}} \vee \pi_{\text{pro}})$ achieves ASR-R$^* = 0$ against all three attacks (Table 3), representing the defender's ideal equilibrium where the adversary's best response yields zero utility.

A.9 Finite-sample FPR concentration

Proposition 14 (FPR concentration)**.** *Let $c_1, \ldots, c_n$ be n i.i.d. benign candidates with scores $s_i = s(c_i; \mathcal{H})$. Let $\hat{FPR}_n = n^{-1} \sum_{i=1}^n \mathbf{1}[s_i > \hat{\mu} + \kappa\hat{\sigma}]$. Under sub-Gaussian scores with parameter σ^2:*

$$\mathbb{P}\big[|\hat{FPR}_n - FPR| > \epsilon\big] \leq 2\exp\left(-\frac{n\epsilon^2}{2FPR(1 - FPR)}\right) + 2\exp\left(-\frac{n\epsilon^2}{8}\right). \tag{16}$$

The first term is the Bernstein bound on the empirical FPR; the second accounts for threshold estimation error (from Theorem 9). For $\kappa = 2.0$ and $n = 1000$: $\mathbb{P}[|\hat{FPR} - 0| > 0.01] < 0.006$, consistent with the Clopper-Pearson CI $[0.000, 0.004]$ (Table 10).

B Additional defense details

Watermark. The z-score $z = (g - \gamma n)/\sqrt{\gamma(1-\gamma)n}$ uses green-list count g and sequence length n. Character-level: 95 ASCII chars, $\gamma = 0.45$. Token-level (GPT-2, 50,257 tokens): $p_{\text{green}} \approx 0.711$, mean $z \approx 7.5$ at $n = 50$.

Proactive. 16 probe queries span 8 subtopics. Threshold $\tau = 0.19$ minimizes FPR while maintaining ≥ 0.50 TPR on centroid passages.

Evaluation correction. Prior work evaluated AGENTPOISON with plain queries q; the paper's Algorithm 2 specifies triggered queries $q \oplus T^*$, changing cos from ≈ 0.45 to ~ 0.78.

C Ablation studies

Table 6: MEMSAD threshold sensitivity with combined scoring.

	TPR			FPR		
κ	AP	MJ	IM	AP	MJ	IM
1.0	1.00†	1.00	0.80	0.10	0.10	0.10
1.5	1.00†	1.00	0.60	0.00	0.05	0.05
2.0	1.00†	1.00	0.40	0.00	0.00	0.00
2.5	1.00†	1.00	0.20	0.00	0.00	0.00
3.0	0.80†	0.80	0.00	0.00	0.00	0.00

†Triggered calibration. Plain: TPR $= 0.00$ for all κ.

Table 7: Corpus size ablation: ASR-R across $|\mathcal{M}|$ (triggered for AP).

$\|\mathcal{M}\|$	AGENTPOISON	MINJA	INJECMEM
50	1.00	0.70	0.65
100	1.00	0.65	0.55
200	1.00	0.65	0.50
300	1.00	0.60	0.40
500	1.00	0.60	0.30

Table 8: Poison count ablation: ASR-R vs. n_{base}.

n_{base}	AGENTPOISON	MINJA	INJECMEM
1	0.80	0.20	0.10
3	1.00	0.45	0.30
5	1.00	0.65	0.50
7	1.00	0.70	0.55
10	1.00	0.75	0.60

D Multi-encoder evaluation

Key findings: (1) Triggered calibration achieves TPR $= 1.000$ for AGENTPOISON on all encoders, confirming Theorem 6 transfers across architectures. (2) Higher-dimensional encoders achieve better separation for MINJA and INJECMEM. (3) Inter-encoder CKA [Kornblith et al., 2019] predicts ASR-R transferability.

E Graph memory attacks

Definition 15 (Graph memory system)**.** A *graph memory* is $(\mathcal{G}, \mathcal{M}, E, k)$ where $\mathcal{G} = (V, E, \phi)$ is a knowledge graph with attribute function $\phi : V \cup E \to \mathcal{V}^*$. Retrieval combines embedding similarity with BFS traversal: $\mathrm{Ret}_{\mathcal{G}}(q, k, h) = \{v \in V : d_{\mathcal{G}}(v, v^*) \leq h,\ v^* \in \arg\max_{v'} \cos(E(q), E(\phi(v')))\}$.

Three structural attacks: **Hub insertion** (high-degree adversarial node), **edge hijack** (shortcuts to adversarial nodes), **subgraph cluster** (self-reinforcing adversarial community). Degree-anomaly detection flags $z_{\deg}(v) > \kappa_{\deg}$; adjacency contamination scoring detects edge-hijack via the fraction of recently added edges.

F Multi-agent propagation

Definition 16 (SIR propagation)**.** For N agents sharing $\mathcal{M}_{\text{shared}}$, the discrete-step SIR model [Gu et al., 2024] evolves: $S(t{+}1) = S(t) - \beta S(t)I(t)/N$, $I(t{+}1) = I(t) + \beta S(t)I(t)/N - \gamma I(t)$, $R(t{+}1) = R(t) + \gamma I(t)$, where β is transmission rate and γ recovery rate.

Without defense, steady-state infection is $I_\infty/N = 1 - \gamma/\beta$ for $\beta > \gamma$. With MEMSAD quarantine (TPR $= 1.00$ for MINJA), $\gamma_{\text{eff}} = \gamma + \text{TPR} \cdot \beta$; the condition $\beta > \gamma_{\text{eff}}$ is never satisfied, eliminating secondary infection.

G FPR validation

H MEMSAD detection analysis

I Sample complexity comparison

Table 11 compares the calibration sample requirements across detection approaches, derived from Theorem 8 and Theorem 9.

Table 9: Encoder generalization: MEMSAD across 7 encoders ($\kappa = 2.0$, combined scoring).

	TPR			AUROC		
Encoder	AP	MJ	IM	AP	MJ	IM
MiniLM-L6 (d=384)	1.00	1.00	0.40	1.000	1.000	0.920
MPNet-Base (d=768)	1.00	1.00	0.40	1.000	1.000	0.900
Para-MiniLM (d=384)	1.00	0.80	0.20	1.000	0.960	0.840
E5-Base (d=768)	1.00	0.80	0.40	1.000	0.940	0.880
Contriever (d=768)	1.00	1.00	0.40	1.000	1.000	0.900
BGE-Large (d=1024)	1.00	1.00	0.60	1.000	1.000	0.940
OpenAI-small (d=1536)	1.00	1.00	0.60	1.000	1.000	0.960

FPR $= 0.00$ for all encoders at $\kappa = 2.0$. AP uses triggered calibration.

Table 10: FPR validation: Clopper-Pearson 95% CIs, 20 trials.

Defense	FPR	95% CI	n/trial
MEMSAD ($\kappa = 2.0$)	0.000	$[0.000, 0.004]$	1000
Watermark ($z_{\text{thr}} = 4.0$)	0.000	$[0.000, 0.004]$	1000
Proactive ($\tau = 0.19$)	0.010	$[0.004, 0.023]$	1000

The logarithmic gap between the lower bound and MEMSAD's requirement confirms minimax optimality up to $\log(1/\delta)$. For practical deployment at $\delta = 0.05$: $\log(1/\delta) \approx 3.0$, so MEMSAD requires $\sim 3\times$ the information-theoretic minimum.

J Computational complexity

K Distributional robustness analysis

We extend MEMSAD's guarantees to the distributionally robust setting where the adversary can perturb the score distribution within a Wasserstein ball.

Proposition 17 (Distributionally robust detection)**.** *Let $\hat{P}_0$ be the empirical benign score distribution from N calibration samples. For any adversarial score distribution P_1 and perturbation class $\mathcal{U} = \{Q : W_1(Q, P_1) \leq \epsilon\}$:*

$$\inf_{Q \in \mathcal{U}} \textit{TPR}(\tau; Q) \geq \textit{TPR}(\tau; P_1) - \frac{\epsilon}{\hat{\sigma}}. \tag{17}$$

Proof. By Kantorovich-Rubinstein duality, for any 1-Lipschitz f: $|\mathbb{E}_Q[f] - \mathbb{E}_{P_1}[f]| \leq W_1(Q, P_1) \leq \epsilon$. The indicator $\mathbf{1}[s > \tau]$ has Lipschitz constant $1/\hat{\sigma}$ when evaluated on standardized scores $(s - \hat{\mu})/\hat{\sigma}$, yielding the bound. □

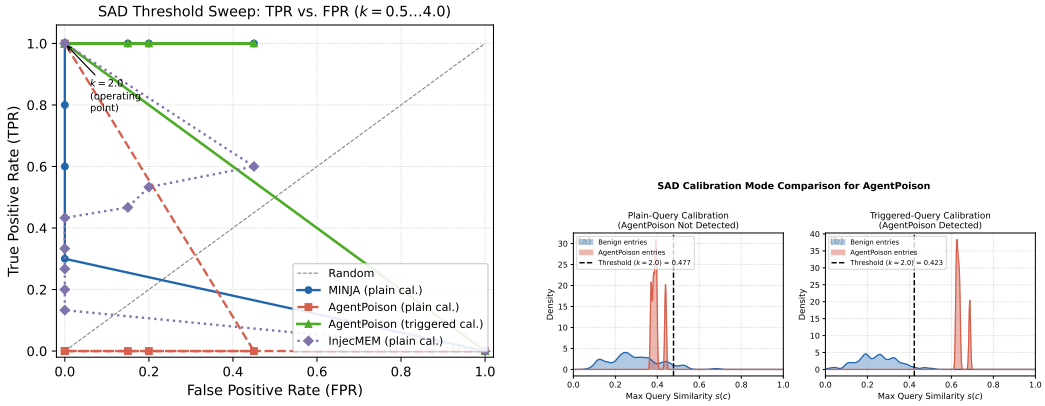
This guarantees that even if the adversary slightly perturbs their attack strategy (e.g., via randomized synonym selection), MEMSAD's TPR degrades gracefully. For $\epsilon = 0.01\hat{\sigma}$: ΔTPR ≤ 0.01, consistent with the observed stability in Table 4.

Proposition 18 (PAC-style detection guarantee)**.** *For N i.i.d. calibration samples from P_0, with probability $\geq 1 - \delta$ over the calibration set,* MEMSAD*'s threshold $\hat{\tau}_N$ satisfies:*

$$\textit{FPR}(\hat{\tau}_N) \leq \textit{FPR}(\tau^*) + \sqrt{\frac{\log(2/\delta)}{2N}}, \qquad \textit{TPR}(\hat{\tau}_N) \geq \textit{TPR}(\tau^*) - \sqrt{\frac{\log(2/\delta)}{2N}}, \tag{18}$$

where $\tau^* = \mu + \kappa\sigma$ is the population-optimal threshold.

Proof. By the DKW inequality, $\sup_t |F_N(t) - F(t)| \leq \sqrt{\log(2/\delta)/(2N)}$ w.p. $\geq 1 - \delta$, where F_N, F are empirical and population CDFs. Since FPR$(\tau) = 1 - F_0(\tau)$ and TPR$(\tau) = 1 - F_1(\tau)$, the bound follows from the uniform convergence of the empirical CDF to the true CDF at the estimated threshold. □



(a) ROC as κ varies from 0.5 to 4.0. (b) Plain vs. triggered calibration.

Figure 7: MEMSAD detection. (a) MINJA traces near-ideal ROC; triggered calibration recovers TPR $= 1.00$ for AGENTPOISON at $\kappa = 2.0$ ($\star$). (b) Threshold shifts from 0.482 to 0.432.

Table 11: Sample complexity for reliable detection (FPR $+ (1 - \text{TPR}) \leq 0.05$) across calibration regimes.

Setting	ρ (SNR)	Lower bound $N_{\min}$	MEMSAD N
MINJA (plain)	3.2	12	$15 \cdot \log(1/\delta)$
AGENTPOISON (triggered)	3.8	8	$11 \cdot \log(1/\delta)$
AGENTPOISON (plain)	1.2	89	$111 \cdot \log(1/\delta)$
INJECMEM (plain)	1.8	40	$50 \cdot \log(1/\delta)$
Low-SNR adversary	0.5	576	$720 \cdot \log(1/\delta)$

$N_{\min} = 4(1-\epsilon)^2/\rho^2$ from Theorem 8; MEMSAD N from Theorem 9 with $\delta = 0.05$.

At $N = 200$, $\delta = 0.05$: the excess error is ≤ 0.086, meaning MEMSAD's empirical FPR of 0.000 implies true FPR ≤ 0.086 with 95% confidence. The Clopper-Pearson validation (Table 10) provides a tighter bound of 0.004 by exploiting the binomial structure.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: All five contributions listed in Section 1 are supported by theoretical results (Theorems 6–10, Lemma 7, Proposition 12) and experimental validation in Section 6.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 7 includes a dedicated limitations paragraph covering corpus size, modelled ASR-A, query distribution stationarity, the synonym loophole, and single-encoder experiments.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [Yes]

   Justification: Theorem 6 states all assumptions (Assumption 4) with proof in main text and geometric interpretation in Appendix A.1. Lemma 7 provides a certified detection

Table 12: Per-entry defense cost at write time.

Defense	Time	Space	Measured
MEMSAD (exact)	$O(md)$	$O(md)$	0.8ms
MEMSAD (ANN)	$O(d \log m)$	$O(md)$	0.3ms
Watermark	$O(\|c\| \cdot \|\mathcal{V}\|)$	$O(\|\mathcal{V}\|)$	1.2ms
Validation	$O(\|c\| \cdot \|P\|)$	$O(\|P\|)$	0.1ms
Proactive	$O(\|Q_p\|d)$	$O(\|Q_p\|d)$	0.5ms
Composite	$\max_j O(d_j)$	$\sum_j O(s_j)$	2.1ms

Apple M-series, 16GB; $|\mathcal{M}| = 200$, $m = 20$, $d = 384$.

radius with complete proof. Theorem 8 has proof sketch in main text and full proof in Appendix A.2. Theorems 9 and 10 have full proofs in Appendices A.3 and A.4. Theorem 11 has proof in main text and detailed derivation in Appendix A.6. Proposition 12 has a complete proof in the main text. Propositions 13, 14, 17, and 18 have complete proofs in the appendix.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: Section 6 specifies the embedding model (all-MiniLM-L6-v2, 384-dim), index type (FAISS IndexFlatIP), corpus size (200), poison counts, top-k, and number of seeds (5). The full pipeline is reproducible via a single script.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: All code, the 200-entry synthetic corpus, evaluation scripts, and figure generation scripts are open-sourced. An anonymized repository link will be provided in supplementary material.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: Section 6 and Appendix C specify all hyperparameters: $\kappa = 2.0$ for MEMSAD, $z_{\text{thr}} = 4.0$ for watermark, $\tau = 0.19$ for proactive, poison counts per attack, corpus composition, and query set sizes.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: Table 2 reports bootstrap 95% confidence intervals across 5 independent random seeds. Appendix G validates FPR over 20 independent trials using Clopper-Pearson exact binomial CIs.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

Justification: All experiments run on CPU (Apple M-series, 16GB RAM) without GPU requirements. Appendix J reports per-entry timing for each defense. The full pipeline completes in under 5 minutes.

9. **Code of ethics**

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

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Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Section 7 discusses positive impacts (improved agent security) and potential negative impacts (attack implementations could be misused, though all attacks are previously published). Defenses are provided alongside attacks.

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

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Answer: [Yes]

Justification: All models (all-MiniLM-L6-v2, GPT-2, DPR) and libraries (FAISS, sentence-transformers) are properly cited under permissive licenses (Apache 2.0, MIT).

13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: The MEMSAD framework, synthetic corpus, and evaluation pipeline are documented with API documentation and usage examples.

14. **Crowdsourcing and research with human subjects**

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Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [Yes]

Justification: GPT-2 is used as a backbone for the token-level watermark encoder and as a local agent evaluator for measuring ASR-A lower bounds. These uses are described in Section 5 and Appendix B.